DO NOW — RECALL

Linked lists and pointers

Name: _____ Class: _____ Date: _____

1. Push 2, 5, 8 then pop. What remains bottom-to-top?

2. What is the first index of a C# array?

3. What is a base case?

4. Queue A,B,C: dequeue once and enqueue D. What remains?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

2, 5. The top 8 is removed.

2. Model answer

0.

3. Model answer

The stopping condition in recursion.

4. Model answer

B,C,D front-to-rear.

Vocabulary — know these terms

node — A list element containing data and a link.

next pointer — Reference or index of the next node.

head — Reference to the first list node.

null sentinel — Marker such as −1 in an index implementation meaning no next node.

traversal — Visit nodes by following successive links.

NEW LEARNING

Read, annotate and explain

Follow references

head = 0 means the first node has value 10. next[0] initially points at node 1 (20).

Prepare the new node

The new node at index 3 stores 15 and next[3] = 1, preserving the rest of the list.

Change the existing link

Set next[0] = 3, producing the chain 10 → 15 → 20 → 30.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Insert value 7 between 4 and 9 in the linked list 4 → 9 → 12.

Predict

Create node 7 and set its next link to the existing node 9.

Confirm and explain

Point node 4 at node 7. The chain becomes 4 → 7 → 9 → 12.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

In linked list 5 → 10 → 15, what follows node 10?

Question 2

A node with value 12 must be inserted between 7 and 20. State the two link changes.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why does inserting into a linked list avoid shifting array elements but may still take O(n) time?

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A singly linked list always supports direct O(1) access to its twentieth element.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.